



Vivarium

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<https://github.com/AnaSantos05/ICG-Vivarium>

<https://anasantos05.github.io/ICG-Vivarium/>

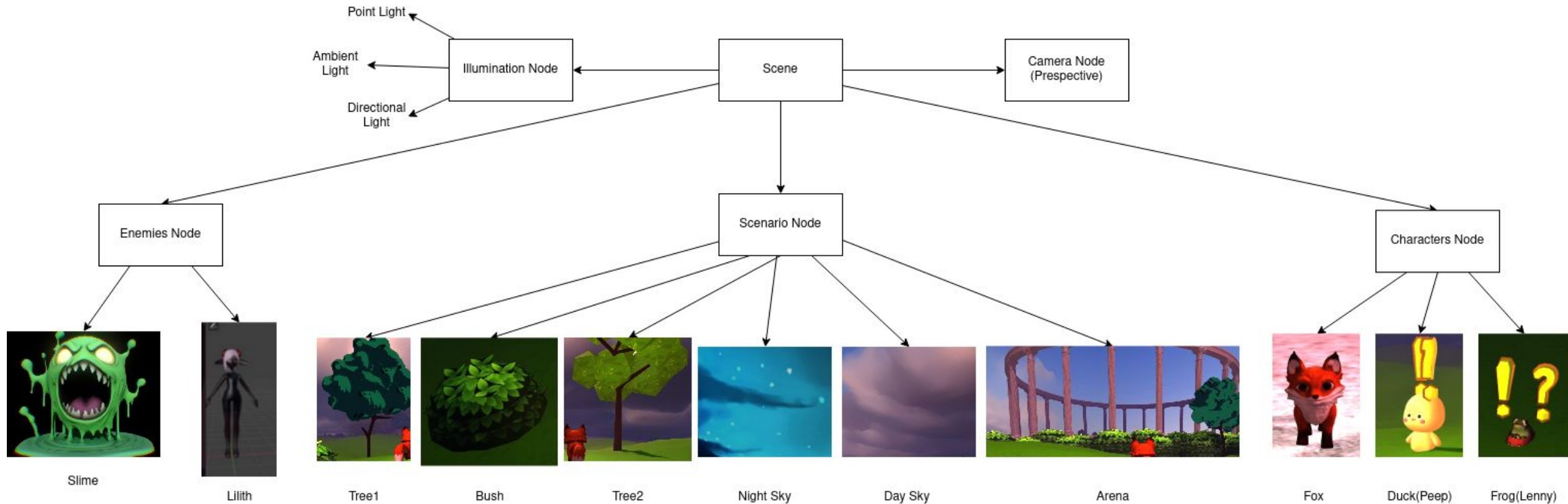
Introduction to Computer Graphics – 2025/2026 – Final Project

The Project

- Initially the project was meant to be a cozy RPG but as I was developing it, I felt I wanted to turn it a little different -> more fight and hero's wit oriented.
- The user can play as a fox, talk to NPCs and hear their story as well as progress in the story by completing their quests and fight bosses.
- Uses the core Three.js module plus additional examples modules (e.g., GLTFLoader, FBXLoader)
- Three.js version: 0.118.1 (CDN import)



Models & Scene Graph



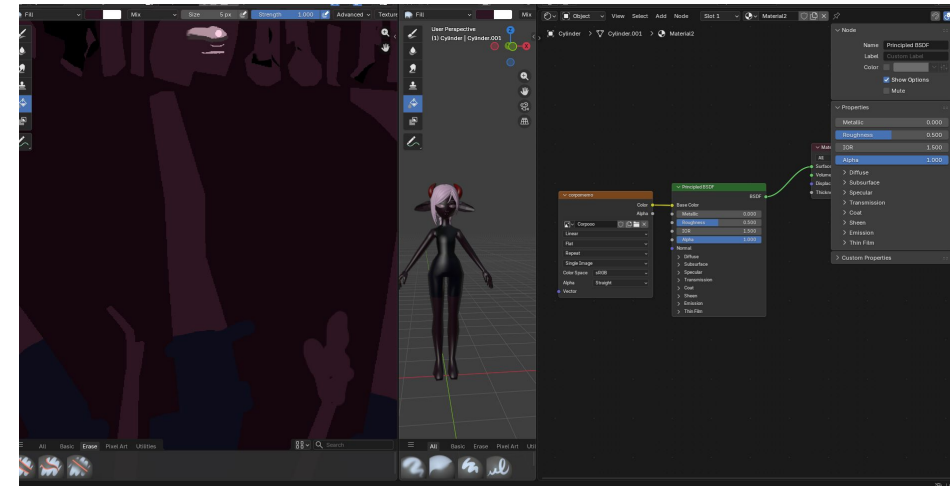
Illumination & Materials & Textures

- **Visual realism:** Height-displaced terrain (sin/cos waves + slight randomness), large tiled ground plane, and a void cover to hide edges. Sky model and real life textures (ex: arena floor).
- **Materials:** MeshStandardMaterial for terrain (physically-based shading), MeshBasicMaterial for the void cover (unlit, simple mask). Image for ground and trees, as well as for characters.
- **Light sources:** Directional light, ambient light and point light.



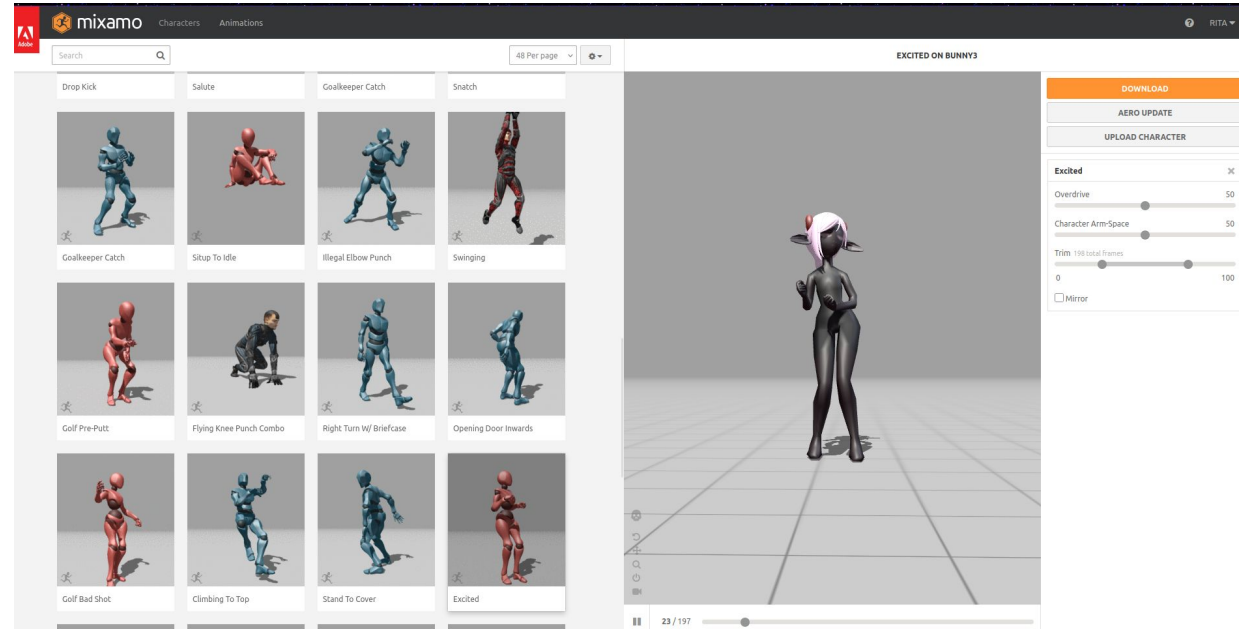
Illumination & Materials & Textures

- **Shaders:** Custom shader patch via `onBeforeCompile` to discard fragments under the arena area
- **Texture maps:** A single color/diffuse map for the grass terrain. UV maps for characters.
- **Purpose of maps:** Provide ground color detail and tiling for visual richness
- **Texture sources:** Loaded from the path in `TERRAIN_CONFIG.texture_path` (in `gameConfig.js`)



Animation

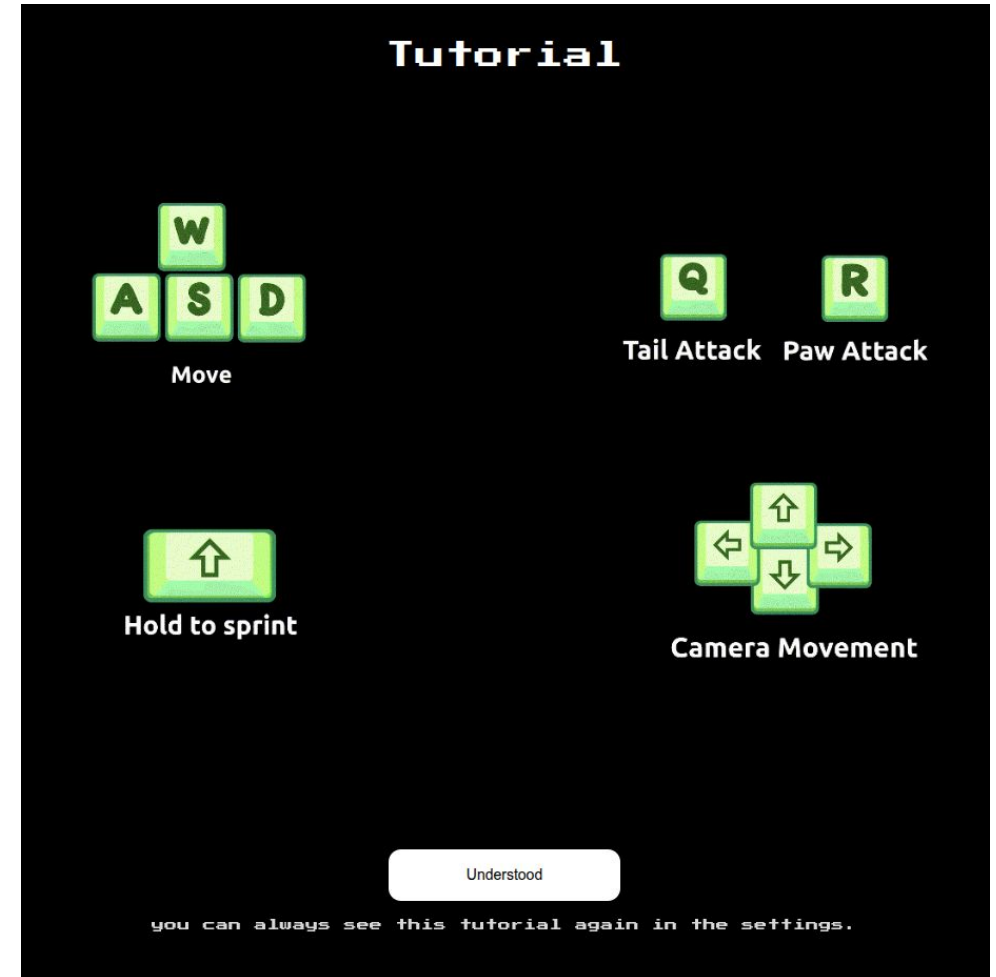
- **Animation for the start of the game (fox moving):** I block the player's control and make it move forward. I change the camera's frustum slowly to the front of the character through literal changes in position.



- **Boss animation:** lilith -> I rigged the skelethon and used Mixamo to put some animations on her.
- **Fox animation:** The fox model already had animations, I used those.
- **Power animations:** I created very simple 'power' looking effects and I moved them with mathematical expressions.

User Interaction

- What kind of camera are you using?
Why?
 - Perspective, third-person orbit camera that follows the player. Supports exploration and combat.
- How is the user interacting with the application?
 - Keyboard-driven movement and mouse for camera(optional)
- What to do: Accept quests, defeat monsters and explore the world



User Interaction

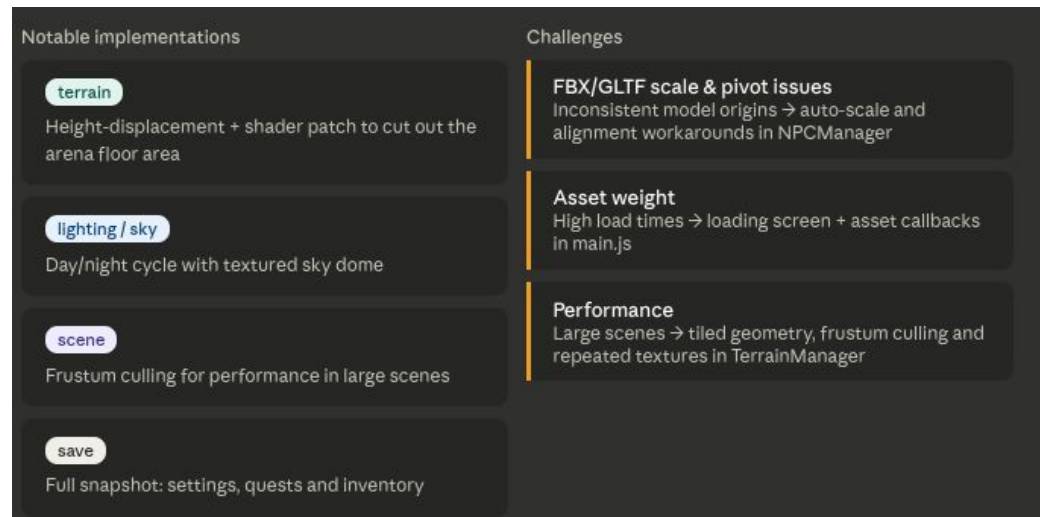
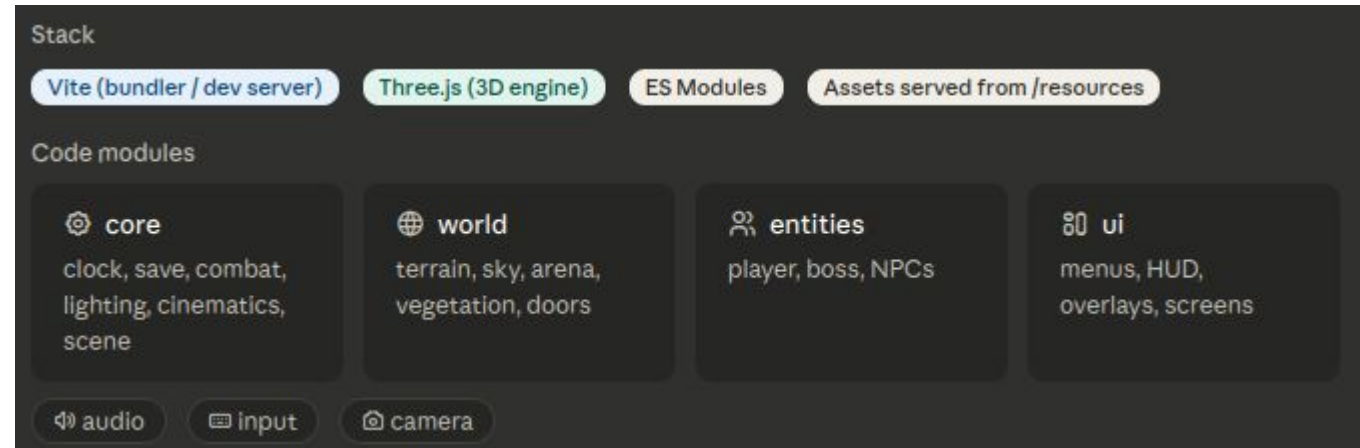
- The user can use the mouse to rotate the camera
- The keyboard is the main control for the game. WASD moved the character, arrows moves the camera.
- Main Menu + Settings + Credits



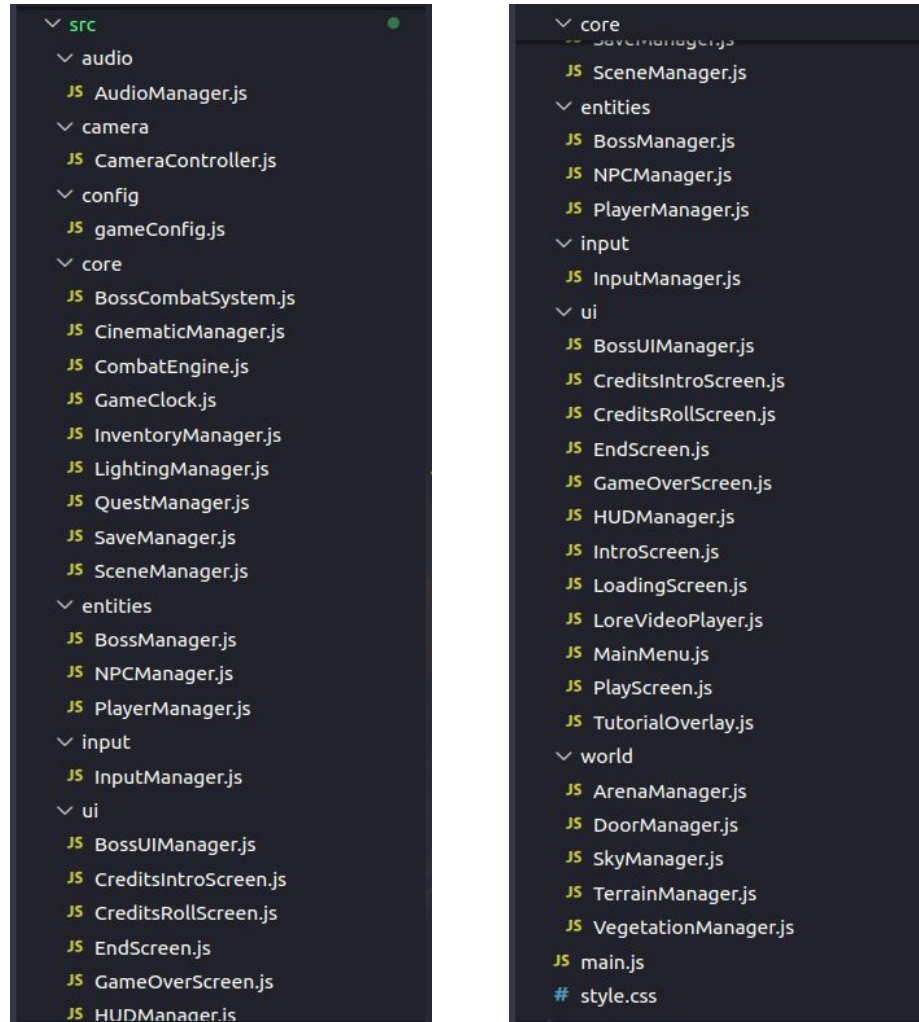
Development



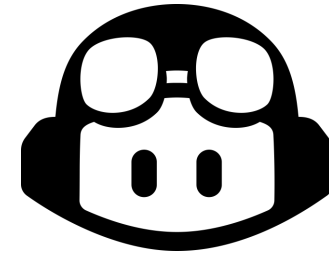
Vite



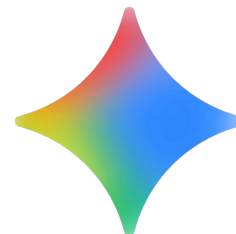
Development



Use of AI



- I used GitHub Copilot to remove code trash from my game (code that I was not using, that was redundant and that I left there as a thinking process)
- I looked ideas and strategies on the internet but there were ones that I didn't quite find how to do. Eg: How to put the arena base in a wavy ground, so I asked Github Copilot to give me solutions and I understood what it did.
- I used Gemini to debug Blender things, such as finding tools or why my textures were not being loaded.



Performance & Conclusions

- **Performance:** Low poly models + frustum culling technique.
- The project was tested in different computers with different performances and different browsers, it works well. No support for mobile so far (because of the controls), but with a keyboard and a mouse, it works.
- **Most interesting:** The story, the quests and the boss fights.
- **What would I do differently:** Adapt it to mobile and doing more research before.

Requirements

- A big world with a Minimap on UI. (F)
- The main character - a fox. A few NPCs as quest givers (min 1). And a minimum of 2 bosses. (F)
- Quest giving/accepting/completion system. (F)
- Fight system and animations. (F)
- Inventory system. (F)
- The game has to be optimized to be able to run on a different variety of computers. (NF)
- A functional tutorial at the beginning of the game. (F)
- Day and night system. (F)
- Ambient music, combat music and SFX. (NF)
- Mini animated story to give context of the game, in the beginning of it. (NF)

Future Work

- Mobile adaptation;
- More cartoony UI;
- More character diversity;
- Better combat engine;
- More sfx;
- Better story development;
- Better accessibility features;

References

- SimonDev. "I Tried Making a 3D RPG Game in JavaScript." YouTube video, 2020.
<https://www.youtube.com/watch?v=SBfZAVzbhCg&t=346s>
- RoBuilder, "ABSOLUTE Beginner Basics.. Blender For Noobs 2025," YouTube video, 2025, <http://www.youtube.com/watch?v=peSv5IT5Ve4>
- BlenderVitals, "Create a Low Poly Tree in Blender in 1 Minute!," YouTube video, 2023, http://www.youtube.com/watch?v=hvxoAX_pol0
- Eve Sculpts, "Blender Tutorial - 2D Drawing to 3D Model (Part 1)," YouTube video, 2021, <http://www.youtube.com/watch?v=AIPPYkZg9D4>
- Classes' materials

Live-Demo

Newer stuff after presentation:

- More responsiveness:
The game looks better in other screen sizes.
- Quick bug fixes

